1 Measurements *a*, *b* and *c* are taken to determine the quantity *x* which is given by

$$x = \frac{ab^2}{c}.$$

The total uncertainty of *x* is calculated to be 12 %.

Which individual percentage uncertainties of *a*, *b* and *c* when combined give this total of 12 %?

	a / %	*b* / %	*c* / %
A	4	2	4
B	2	8	2
C	3	2	2
D	6	2	4

L2 **Answer: A**

$x = ab^2/c$

$\Delta x/x = \Delta a/a + 2\Delta b/b + \Delta c/c$

$(\Delta x/x) \times 100\% = (\Delta a/a) \times 100\% + 2(\Delta b/b) \times 100\% + (\Delta c/c) \times 100\%$

	a / %	*b* / %	*c* / %	$(\Delta a/a) \times 100\% + 2(\Delta b/b) \times 100\% + (\Delta c/c) \times 100\%$
A	4	2	4	**4 + 2(2) + 4 = 12%**
B	2	8	2	$2 + 2(8) + 2 \neq 12\%$
C	3	2	2	$3 + 2(2) + 2 \neq 12\%$
D	6	2	4	$6 + 2(2) + 4 \neq 12\%$